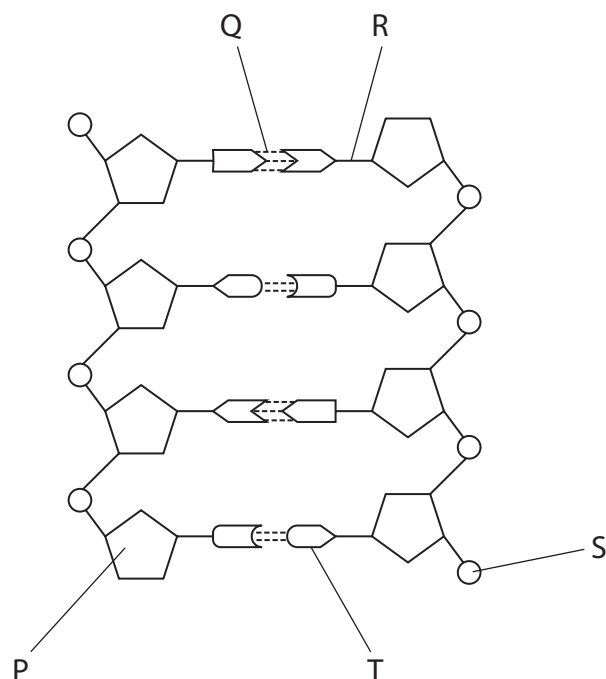


- 7 Analysis of DNA samples taken from endangered species can be used in the conservation and management of these species.

(a) The diagram shows part of a molecule of DNA.



- (i) Which row of the table identifies the bonds **Q** and **R**?

(1)

	Bond Q	Bond R
☐ A	covalent	hydrogen
☐ B	hydrogen	covalent
☐ C	hydrogen	phosphodiester
☐ D	phosphodiester	covalent

- (ii) Which is a negatively charged part of a DNA molecule?

(1)

- ☐ **A** P
☐ **B** R
☐ **C** S
☐ **D** T

(b) The Maleo is an endangered bird endemic to Indonesia.

The photograph shows a Maleo, digging a hole in sand to lay her egg.



(Source: © Avalon.red/Alamy Stock Photo)

Maleos live in the forest and only come to the beaches to lay their eggs. Once the eggs are laid and buried in the sand, the birds return to the forest.

When the egg hatches the young bird digs its way out of the sand and can fly straightaway.

Habitat fragmentation is thought to be partly responsible for this species being endangered. The forests are separated from the beaches by the activities of humans.

(i) State the meaning of the term **endemic**, as used in the context of this question.

(1)

(ii) Give **one** reason why habitat fragmentation could endanger these birds.

(1)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(iii) Suggest the advantages of:

- laying the egg in the sand
- the adults leaving the site
- the young birds being able to fly.

(3)

(iv) In a genetic study of these birds, DNA from the developing chick was isolated from the remains of the eggshell and amplified using PCR.

To amplify a DNA molecule using PCR, two different primers are needed.

Explain why two different primers are needed to amplify one molecule of DNA in a PCR.

(2)

- (v) These birds are thought to be monogamous, which means that they only mate with one bird of the opposite sex.

Birds are only monogamous if the ratio of male to female birds is 1 : 1.

In one genetic study, DNA from 23 eggshells was analysed.

This study found that 8 of the chicks were female and 15 chicks were male.

Comment on the claim that Maleos are monogamous.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for Question 7 = 12 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

